

CODE SPORT



In this month's column, we will discuss some design concepts in Natural Language Processing (NLP).

We have discussed coding questions in the last couple of columns. A few of our readers had reached out to us asking that we also discuss ML/NLP system design questions in our articles, since these are frequently asked as part of ML/NLP interviews in almost all the major companies. In this month's column, we are going to discuss a few popular ML/NLP system design interview questions. We will discuss a couple of questions in detail first, followed by a set of questions for our readers to practice with.

Unlike coding interview questions, design interviews don't have a specific answer/solution. These are intended to test how you would consider multiple design choices to solve a particular problem, select a particular choice based on a certain rationale that you are able to articulate, and in the process, display your deep technical knowledge of the area. Hence, the question framing in design interview questions is generally sketchy. You are expected to ask follow-up questions to get the required information needed to design the system completely. Hence, system design interviews are typically interactive, with the interviewee clarifying his/her assumptions and seeking additional information, especially during the early phase of the interview.

Once you have the needed information, you should sketch out the external interfaces of the system first. These would typically be the APIs that your system would use to interact with the external world. It would be helpful to draw a high-level block diagram indicating key components of your system and enumerating the different choices for each of these components along with their pros/cons. Once

you settle on a specific set of component choices, you should describe their internals/algorithms and if and how these can be optimised. While the above would cover any general system design interview question, for ML/NLP system design questions, it is important to clearly enunciate the points around data availability, quality of data, quality of annotated/labelled data, metrics needed both for intrinsic model evaluation and external evaluation, and choice of models with their rationales.

Let us look at a simple system design question... Design a spell-checker system. Given this one-line question, you need to first figure out where it would be used. Is this going to be a part of a word processing system, or a part of an input processing of a Web search engine front-end? The latency requirements would differ drastically in the two, with the first problem being handled by a post-processing offline model, whereas the second will need an online in-line processing model. So, this is the first question you would ask the interviewer. Let us assume that we want to plug this spell-checker at the input of a Web search engine client.

Next, we need to get clarity on how the module's output will be used. You may want to clarify this with a set of questions to the interviewer.

- If the model flags a spelling error, will it automatically correct it proactively, or will it throw a suggestion to the user saying, "Did you mean ... instead?", as it happens with most of the Web search engine clients. Think for a minute on why this choice of auto-correcting vs throwing a suggestion to the user is important in the overall scheme of things. *Clue:* How would you get the feedback for the spell corrections provided by the system?

